Dataset Validation

Surgical Scene Semantic Segmentation

Dataset: cholec_seg8k

Classes: 13 | Videos: 17 | Frames: 8,080

Class presence is analysed at three levels - frames, pixels and videos - on the complete dataset (no sampling), globally and for every split. Video coverage highlights class imbalance that frame and pixel counts alone can hide.

1. Executive Summary

Dataset: cholec_seg8k

Dataset

- 13 classes, 17 videos, 8,080 frames.
- train: 5,360 frames, 11 videos
- validation: 2,720 frames, 6 videos
- Ignored (out-of-range) pixels: 0

Class imbalance warnings (global, thresholds: <= 3 videos, >= 50% pixels in one video)

- Blood: in 1/17 videos (6%), 100% of its pixels in video01
- Hepatic Vein: in 1/17 videos (6%), 100% of its pixels in video12
- Liver Ligament: in 1/17 videos (6%), 100% of its pixels in video09
- Connective Tissue: in 2/17 videos (12%), 70% of its pixels in video01
- Cystic Duct: in 3/17 videos (18%), 95% of its pixels in video01

Classes missing from a split

- validation: Connective Tissue, Blood, Hepatic Vein, Liver Ligament

Dominant / smallest pixel share (global)

- Dominant: Background (27.24%)
- Smallest: Hepatic Vein (0.0095%)

2. Dataset Overview

Dataset: cholec_seg8k

Split	Frames	Videos
train	5,360	11
validation	2,720	6
global	8,080	17

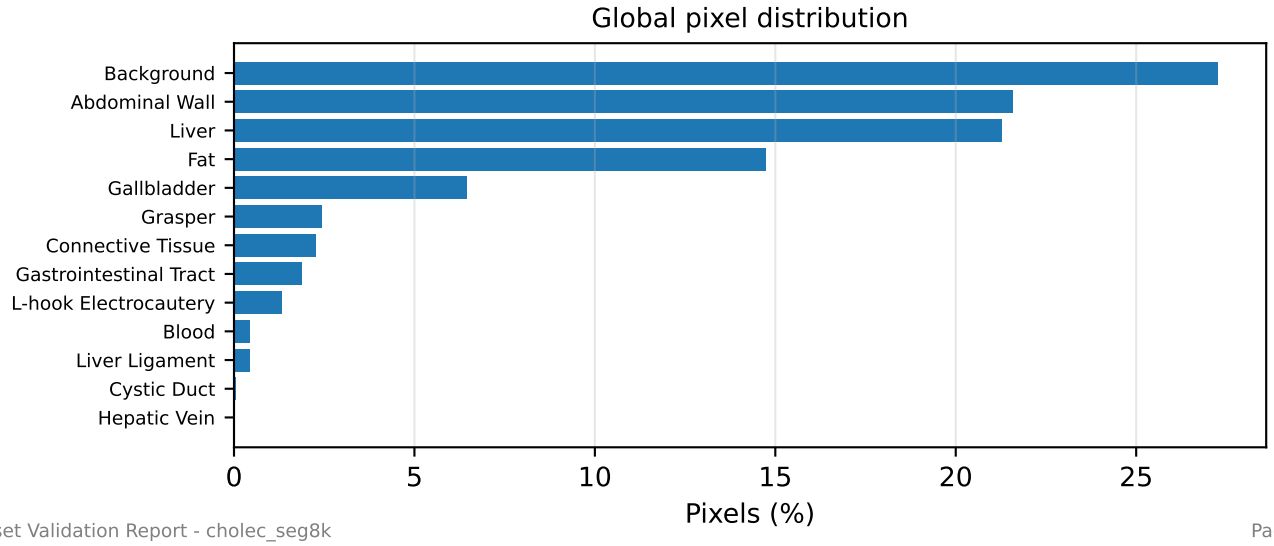
Classes

```
 0  Background
 1  Abdominal Wall
 2  Liver
 3  Gastrointestinal Tract
 4  Fat
 5  Grasper
 6  Connective Tissue
 7  Blood
 8  Cystic Duct
 9  L-hook Electrocautery
10  Gallbladder
11  Hepatic Vein
12  Liver Ligament
```

3. Global Statistics

All splits together - rows highlighted = class imbalance warning (part 1)

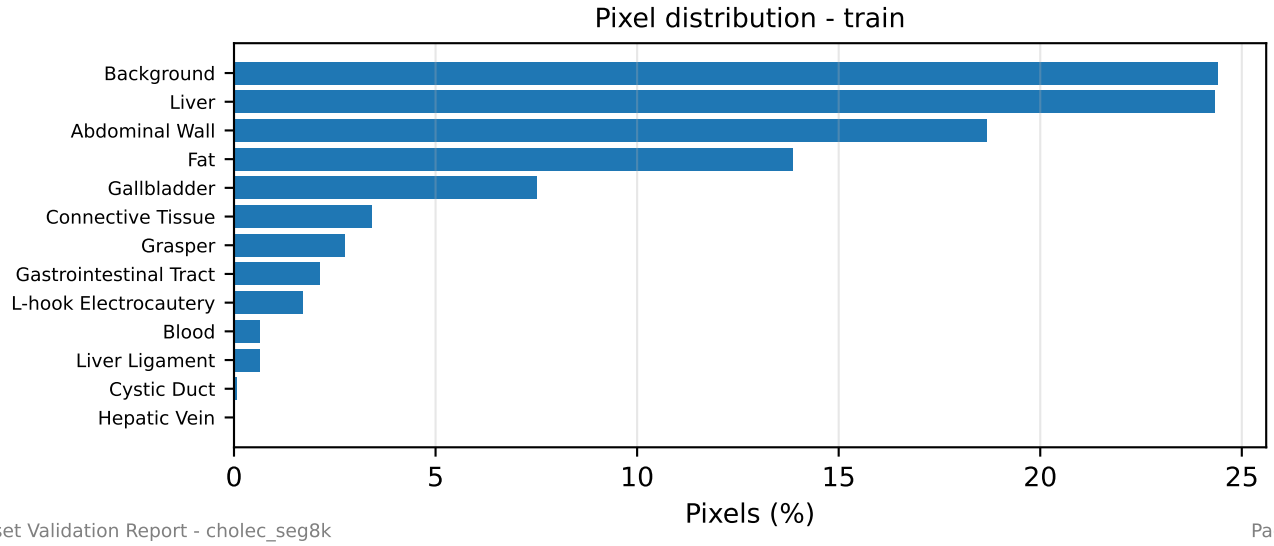
Class	Frames	Frames %	Pixels	Pixels %	Videos
Background	8,080	100.00	902,257,162	27.2408	17/17
Abdominal Wall	7,255	89.79	714,398,034	21.5690	16/17
Liver	8,080	100.00	704,539,925	21.2714	17/17
Gastrointestinal Tract	4,558	56.41	62,289,870	1.8806	13/17
Fat	7,510	92.95	487,565,358	14.7205	17/17
Grasper	6,020	74.50	80,214,947	2.4218	16/17
Connective Tissue	1,600	19.80	75,080,239	2.2668	2/17
Blood	692	8.56	13,991,063	0.4224	1/17
Cystic Duct	248	3.07	1,222,211	0.0369	3/17
L-hook Electrocautery	2,254	27.90	43,433,062	1.3113	9/17
Gallbladder	6,861	84.91	212,978,145	6.4302	16/17
Hepatic Vein	317	3.92	314,317	0.0095	1/17
Liver Ligament	240	2.97	13,869,267	0.4187	1/17



4. Split Statistics - train

5,360 frames, 11 videos - highlighted = class in <= 3 videos (part 1)

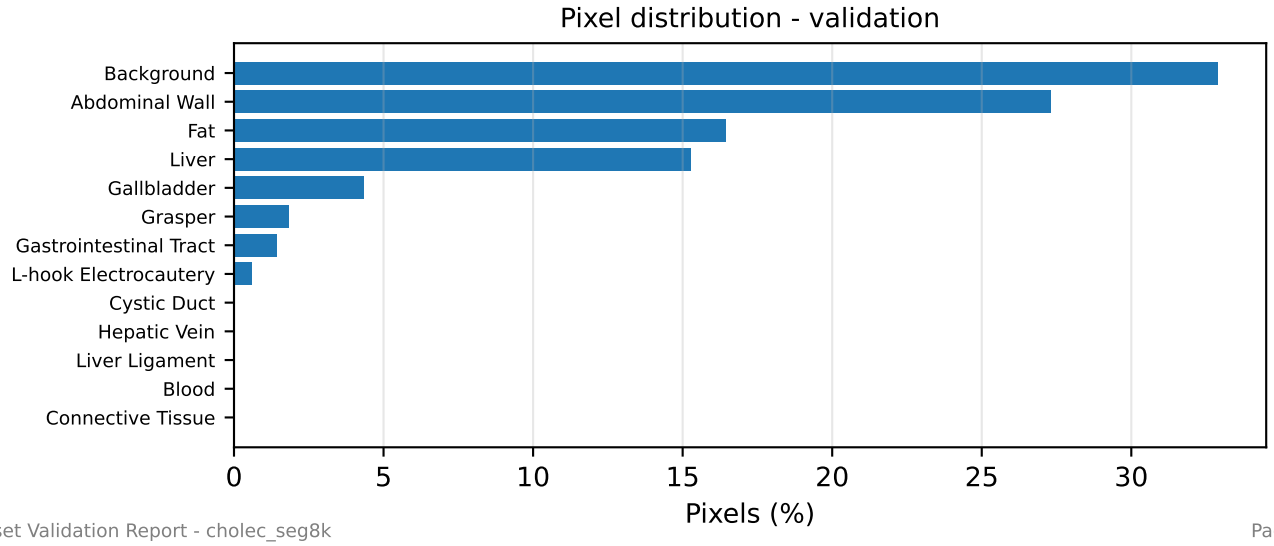
Class	Frames	Frames %	Pixels	Pixels %	Videos
Background	5,360	100.00	535,797,674	24.3858	11/11
Abdominal Wall	4,624	86.27	410,013,131	18.6610	10/11
Liver	5,360	100.00	534,385,476	24.3215	11/11
Gastrointestinal Tract	2,746	51.23	46,409,566	2.1122	8/11
Fat	5,039	94.01	304,187,615	13.8445	11/11
Grasper	4,158	77.57	60,016,433	2.7315	10/11
Connective Tissue	1,600	29.85	75,080,239	3.4171	2/11
Blood	692	12.91	13,991,063	0.6368	1/11
Cystic Duct	246	4.59	1,155,394	0.0526	2/11
L-hook Electrocautery	2,089	38.97	37,084,296	1.6878	6/11
Gallbladder	4,436	82.76	164,866,729	7.5036	10/11
Hepatic Vein	317	5.91	314,317	0.0143	1/11
Liver Ligament	240	4.48	13,869,267	0.6312	1/11



4. Split Statistics - validation

2,720 frames, 6 videos - highlighted = class in <= 3 videos (part 1)

Class	Frames	Frames %	Pixels	Pixels %	Videos
Background	2,720	100.00	366,459,488	32.8668	6/6
Abdominal Wall	2,631	96.73	304,384,903	27.2995	6/6
Liver	2,720	100.00	170,154,449	15.2607	6/6
Gastrointestinal Tract	1,812	66.62	15,880,304	1.4243	5/6
Fat	2,471	90.85	183,377,743	16.4467	6/6
Grasper	1,862	68.46	20,198,514	1.8116	6/6
Connective Tissue	0	0.00	0	0.0000	0/6
Blood	0	0.00	0	0.0000	0/6
Cystic Duct	2	0.07	66,817	0.0060	1/6
L-hook Electrocautery	165	6.07	6,348,766	0.5694	3/6
Gallbladder	2,425	89.15	48,111,416	4.3150	6/6
Hepatic Vein	0	0.00	0	0.0000	0/6
Liver Ligament	0	0.00	0	0.0000	0/6



5. Video Coverage

In how many (and which) videos each class appears - global. Top share = % of the class pixels in its dominant video

Class	Videos	Videos %	Top video	Top share %	Which videos
Background	17/17	100	video24	18	video01, video09, video12, video17, video18, ... (+12)
Abdominal Wall	16/17	94	video52	17	video01, video09, video12, video17, video18, ... (+11)
Liver	17/17	100	video01	30	video01, video09, video12, video17, video18, ... (+12)
Gastrointestinal Tract	13/17	76	video24	16	video01, video09, video12, video17, video18, ... (+8)
Fat	17/17	100	video43	12	video01, video09, video12, video17, video18, ... (+12)
Grasper	16/17	94	video28	18	video01, video12, video17, video18, video20, ... (+11)
Connective Tissue	2/17	12	video01	70	video01, video12
Blood	1/17	6	video01	100	video01
Cystic Duct	3/17	18	video01	95	video01, video12, video17
L-hook Electrocautery	9/17	53	video01	40	video01, video12, video17, video18, video20, ... (+4)
Gallbladder	16/17	94	video01	28	video01, video12, video17, video18, video20, ... (+11)
Hepatic Vein	1/17	6	video12	100	video12
Liver Ligament	1/17	6	video09	100	video09

5. Video Coverage - train

11 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
Background	11/11	video01	18	video01, video09, video12, video20, video26, ... (+6)
Abdominal Wall	10/11	video52	29	video01, video09, video12, video20, video26, ... (+5)
Liver	11/11	video01	39	video01, video09, video12, video20, video26, ... (+6)
Gastrointestinal Tract	8/11	video55	21	video01, video09, video12, video20, video26, ... (+3)
Fat	11/11	video28	19	video01, video09, video12, video20, video26, ... (+6)
Grasper	10/11	video28	23	video01, video12, video20, video26, video27, ... (+5)
Connective Tissue	2/11	video01	70	video01, video12
Blood	1/11	video01	100	video01
Cystic Duct	2/11	video01	100	video01, video12
L-hook Electrocautery	6/11	video01	47	video01, video12, video20, video37, video52, ... (+1)
Gallbladder	10/11	video01	36	video01, video12, video20, video26, video27, ... (+5)
Hepatic Vein	1/11	video12	100	video12
Liver Ligament	1/11	video09	100	video09

5. Video Coverage - validation

6 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
Background	6/6	video24	44	video17, video18, video24, video25, video43, ... (+1)
Abdominal Wall	6/6	video43	37	video17, video18, video24, video25, video43, ... (+1)
Liver	6/6	video24	36	video17, video18, video24, video25, video43, ... (+1)
Gastrointestinal Tract	5/6	video24	64	video17, video18, video24, video25, video48
Fat	6/6	video43	33	video17, video18, video24, video25, video43, ... (+1)
Grasper	6/6	video25	28	video17, video18, video24, video25, video43, ... (+1)
Connective Tissue	0/6	-	0	-
Blood	0/6	-	0	-
Cystic Duct	1/6	video17	100	video17
L-hook Electrocautery	3/6	video18	100	video17, video18, video48
Gallbladder	6/6	video24	25	video17, video18, video24, video25, video43, ... (+1)
Hepatic Vein	0/6	-	0	-
Liver Ligament	0/6	-	0	-

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 1/3, videos part 1/1)

	Background		Abdominal Wall		Liver		Gastrointestinal Tract		Fat		Grasper	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video01 [train]	1,280	94.18M	802	29.15M	1,280	210.24M	381	8.11M	960	28.65M	849	9.23M
video09 [train]	240	16.46M	240	26.79M	240	18.37M	240	6.82M	240	16.08M	0	0
video12 [train]	640	91.79M	639	32.53M	640	54.19M	556	4.18M	640	36.33M	577	4.35M
video20 [train]	160	7.40M	160	9.74M	160	22.91M	112	1.15M	160	9.19M	160	5.92M
video26 [train]	320	40.37M	303	16.35M	320	41.85M	317	9.77M	319	13.84M	182	1.39M
video27 [train]	400	39.55M	400	65.88M	400	16.02M	400	1.39M	400	34.07M	304	3.39M
video28 [train]	560	35.72M	560	67.12M	560	44.81M	501	5.19M	560	57.43M	326	14.05M
video35 [train]	240	6.40M	0	0	240	50.30M	0	0	240	19.24M	240	5.51M
video37 [train]	480	83.47M	480	35.24M	480	18.79M	0	0	480	32.04M	480	4.72M
video52 [train]	800	91.73M	800	119.54M	800	36.93M	0	0	800	49.89M	800	5.63M
video55 [train]	240	28.72M	240	7.67M	240	19.98M	239	9.80M	240	7.43M	240	5.82M
video17 [validation]	320	30.88M	320	44.18M	320	16.45M	296	2.30M	320	24.61M	240	1.49M
video18 [validation]	160	5.08M	151	1.46M	160	28.13M	79	320.0k	160	11.61M	160	3.11M
video24 [validation]	960	159.59M	880	101.12M	960	61.16M	887	10.20M	711	45.49M	400	3.75M
video25 [validation]	320	45.61M	320	26.57M	320	20.40M	320	1.23M	320	25.08M	320	5.71M
video43 [validation]	720	85.66M	720	113.64M	720	25.71M	0	0	720	60.87M	502	5.00M
video48 [validation]	240	39.64M	240	17.42M	240	18.31M	230	1.84M	240	15.72M	240	1.13M
TOTAL	**8,080**	**902.26M**	**7,255**	**714.40M**	**8,080**	**704.54M**	**4,558**	**62.29M**	**7,510**	**487.57M**	**6,020**	**80.21M**

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 2/3, videos part 1/1)

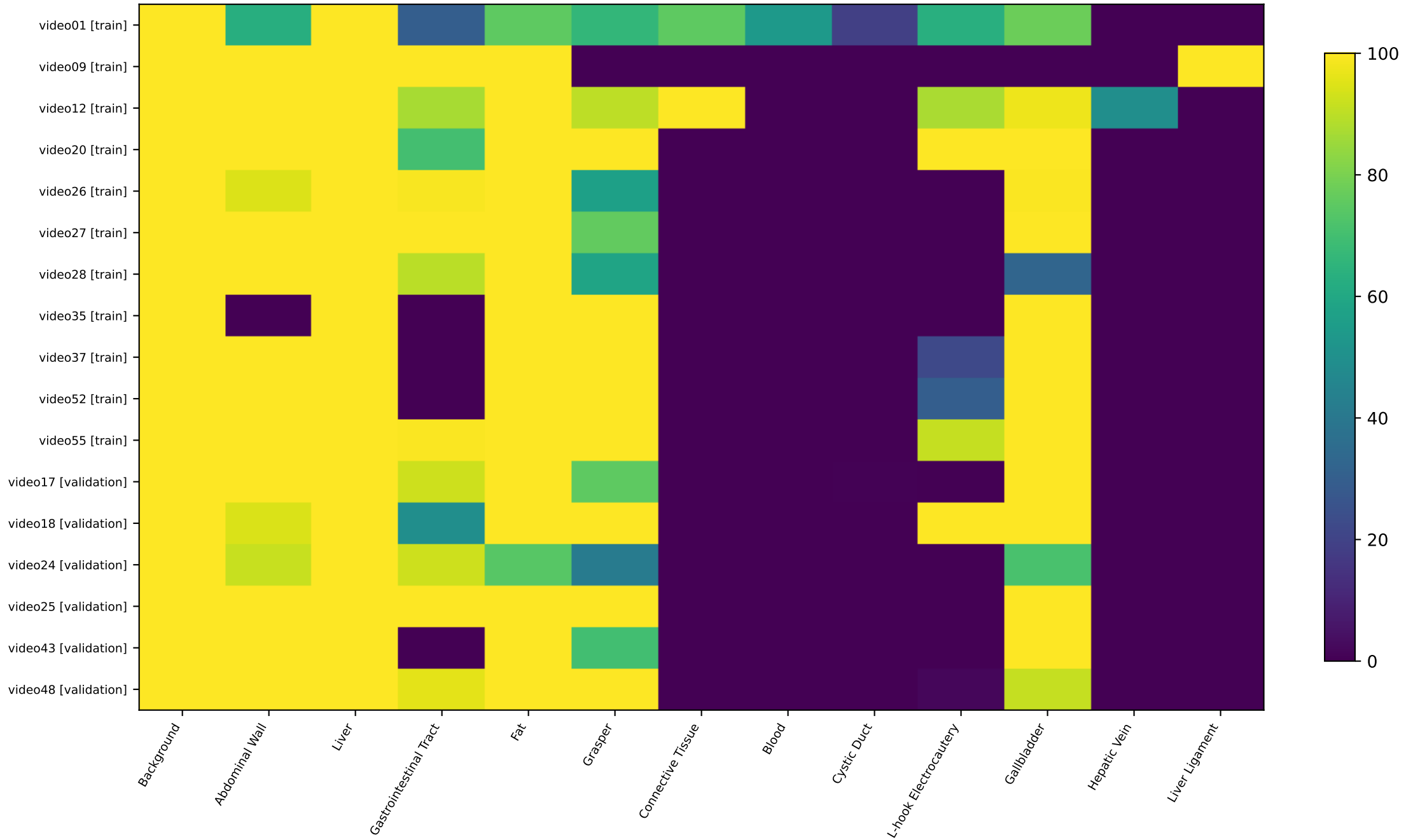
	Connective Tissue		Blood		Cystic Duct		L-hook Electrocautery		Gallbladder		Hepatic Vein	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video01 [train]	960	52.82M	692	13.99M	244	1.16M	806	17.42M	992	59.76M	0	0
video09 [train]	0	0	0	0	0	0	0	0	0	0	0	0
video12 [train]	640	22.26M	0	0	2	3	558	6.72M	624	9.68M	317	314.3k
video20 [train]	0	0	0	0	0	0	160	3.03M	160	6.23M	0	0
video26 [train]	0	0	0	0	0	0	0	0	318	7.61M	0	0
video27 [train]	0	0	0	0	0	0	0	0	400	3.67M	0	0
video28 [train]	0	0	0	0	0	0	0	0	183	5.24M	0	0
video35 [train]	0	0	0	0	0	0	0	0	240	16.91M	0	0
video37 [train]	0	0	0	0	0	0	106	2.11M	479	20.39M	0	0
video52 [train]	0	0	0	0	0	0	240	3.86M	800	20.36M	0	0
video55 [train]	0	0	0	0	0	0	219	3.94M	240	15.01M	0	0
video17 [validation]	0	0	0	0	2	66.8k	1	253	320	11.21M	0	0
video18 [validation]	0	0	0	0	0	0	160	6.35M	160	9.54M	0	0
video24 [validation]	0	0	0	0	0	0	0	0	686	12.21M	0	0
video25 [validation]	0	0	0	0	0	0	0	0	320	6.57M	0	0
video43 [validation]	0	0	0	0	0	0	0	0	720	4.27M	0	0
video48 [validation]	0	0	0	0	0	0	4	306	219	4.32M	0	0
TOTAL	**1,600**	**75.08M**	**692**	**13.99M**	**248**	**1.22M**	**2,254**	**43.43M**	**6,861**	**212.98M**	**317**	**314.3k**

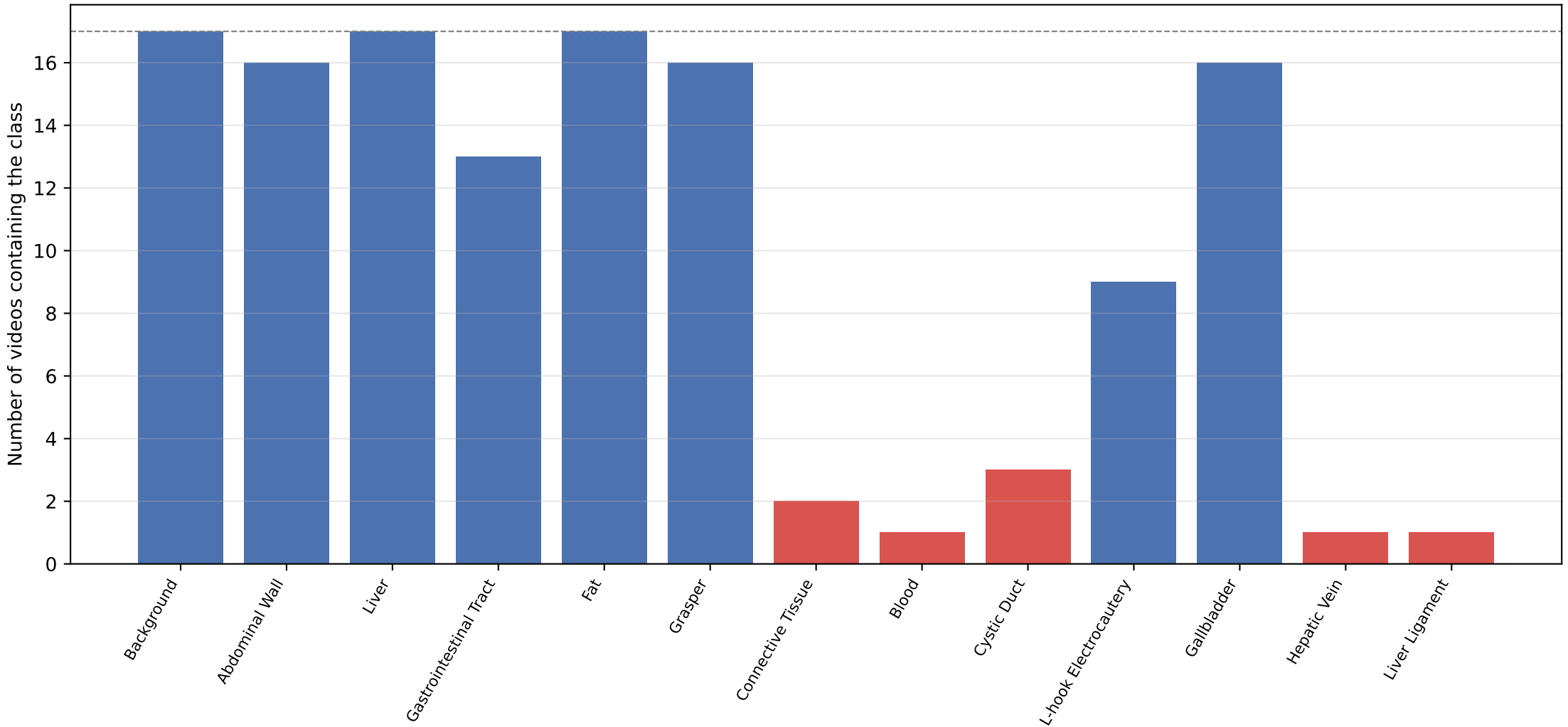
5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 3/3, videos part 1/1)

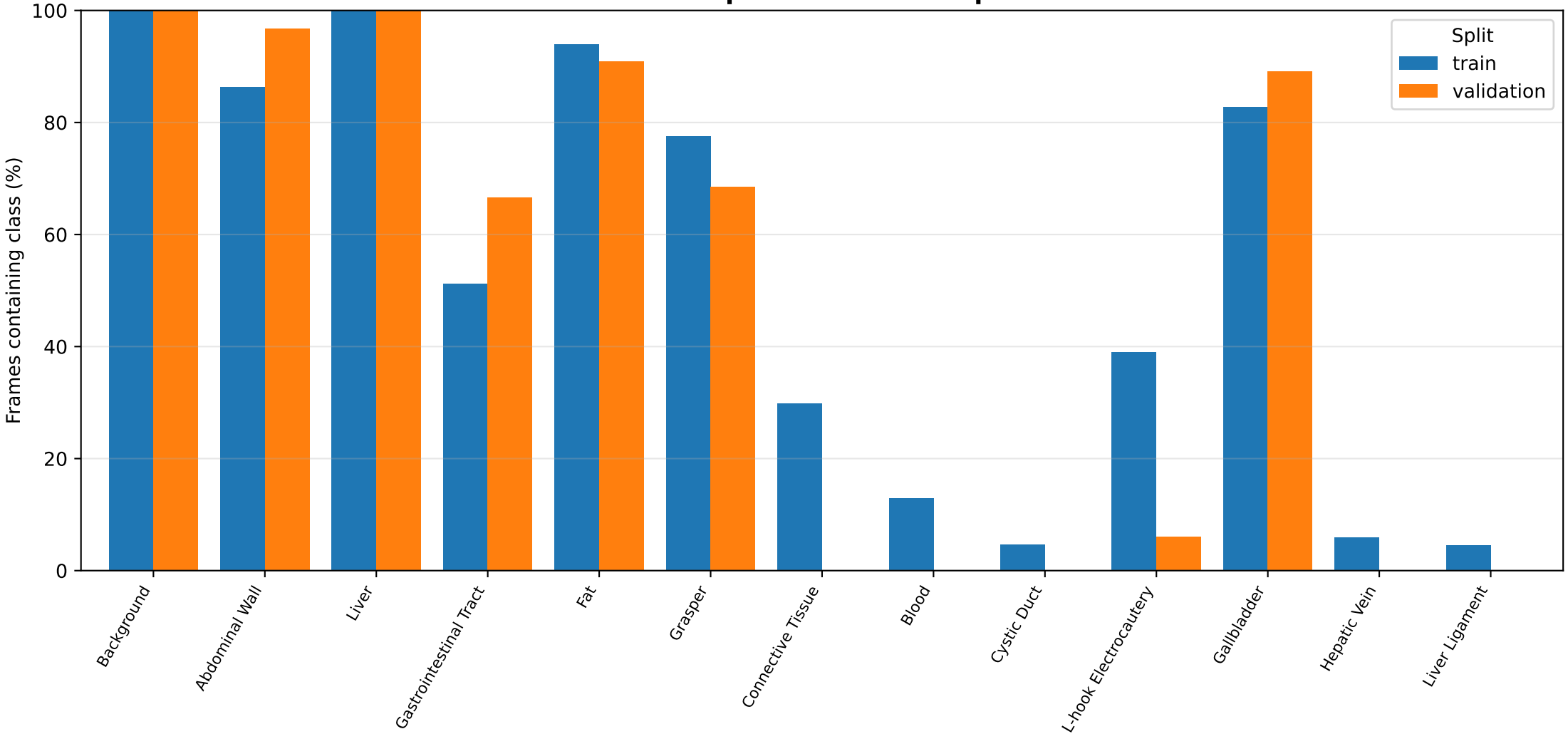
	Liver Ligament	
Video [split]	**frames**	**pixels**
video01 [train]	0	0
video09 [train]	240	13.87M
video12 [train]	0	0
video20 [train]	0	0
video26 [train]	0	0
video27 [train]	0	0
video28 [train]	0	0
video35 [train]	0	0
video37 [train]	0	0
video52 [train]	0	0
video55 [train]	0	0
video17 [validation]	0	0
video18 [validation]	0	0
video24 [validation]	0	0
video25 [validation]	0	0
video43 [validation]	0	0
video48 [validation]	0	0
TOTAL	**240**	**13.87M**

Class presence per video (% of the video's frames containing the class)

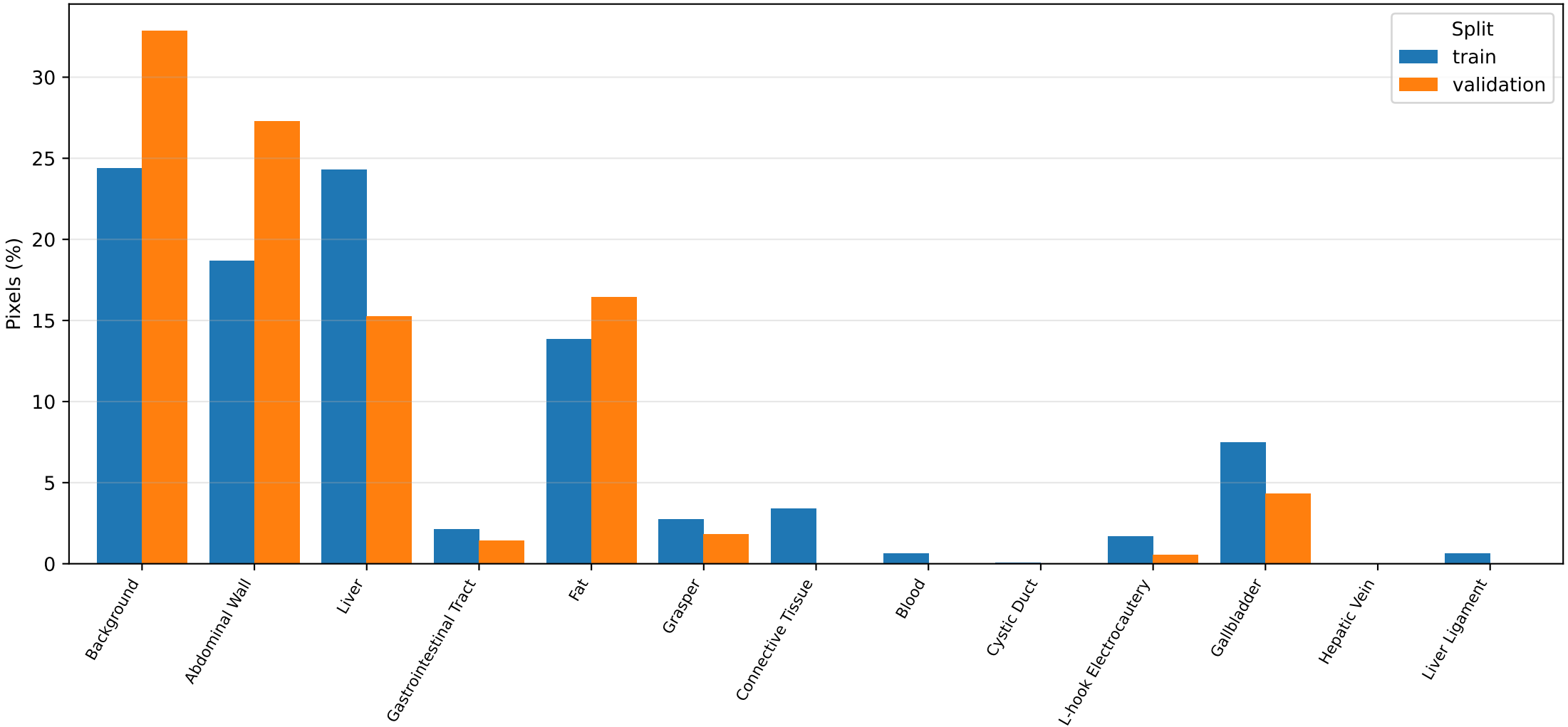



Videos per class (red = imbalance warning, dashed = total videos)
Number of videos containing the class
0
2
4
6
8
10
12
14
16
Background
Abdominal Wall
Liver
Gastrointestinal Tract
Fat
Grasper
Connective Tissue
Blood
Cystic Duct
L-hook Electrocautery
Gallbladder
Hepatic Vein
Liver Ligament

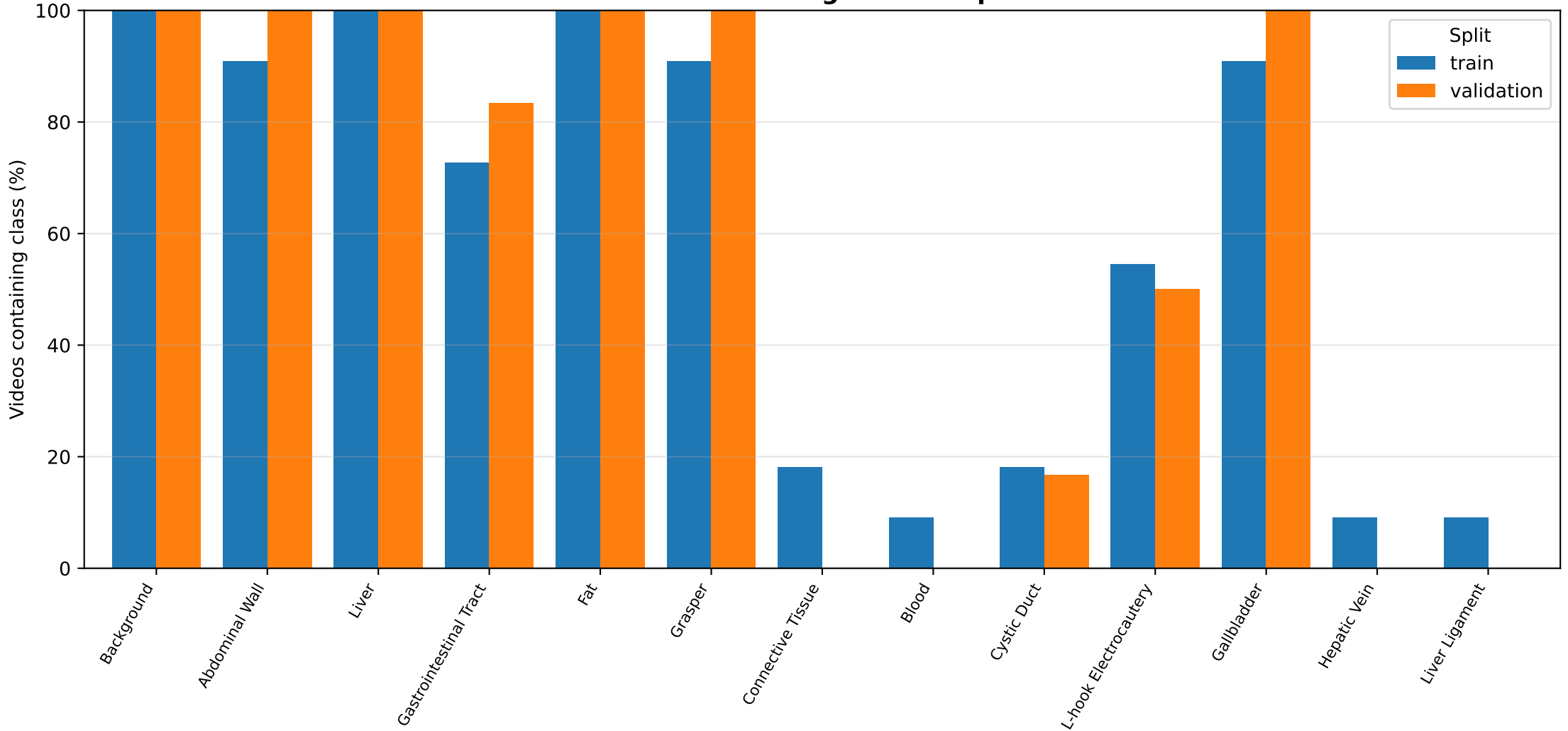
6. Class presence across splits



6. Pixel distribution across splits



6. Video coverage across splits



7. Qualitative Analysis - train

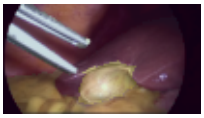
mage - #5266 (video28)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Grasper, Gallbladder

mage - #5905 (video37)



Ground truth

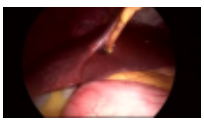


Overlay



Classes: Background, Abdominal Wall, Liver, Fat, Grasper, Gallbladder

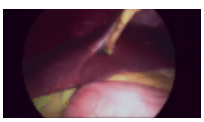
mage - #4080 (video26)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Gallbladder

7. Qualitative Analysis - train

e - #7322 (video52)



Ground truth

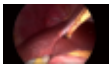


Overlay



Classes: Background, Abdominal Wall, Liver, Fat, Grasper, Gallbladder

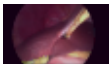
e - #4111 (video26)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Gallbladder

ge - #477 (video01)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Grasper, Connective Tissue, L-hook Electrocautery, Gallbladder

7. Qualitative Analysis - validation

mage - #6098 (video43)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Fat, Gallbladder

mage - #6422 (video43)



Ground truth

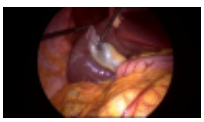


Overlay



Classes: Background, Abdominal Wall, Liver, Fat, Grasper, Gallbladder

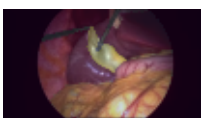
mage - #3497 (video24)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Grasper, Gallbladder

7. Qualitative Analysis - validation

mage - #6655 (video43)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Fat, Grasper, Gallbladder

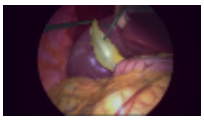
mage - #3512 (video24)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Grasper, Gallbladder

mage - #2402 (video17)



Ground truth



Overlay



Classes: Background, Abdominal Wall, Liver, Fat, Gallbladder

8. Conclusions

Dataset: cholec_seg8k

Frames, pixels and videos measure three different things:
- Frames: how often the class occurs across images.
- Pixels: how much image area the class occupies.
- Videos: how many independent patients/procedures show the class.

A class present in many frames but in only a few videos is a weak basis for generalisation: frames of the same video are highly correlated, so the effective sample size is closer to the number of videos than to the number of frames.

Splits are made at video level, so a rare class can end up entirely in train or entirely in validation/test. Check 'Classes missing from a split' in the executive summary before interpreting per-class metrics.

Classes flagged in red (few videos, or concentrated in one video) deserve special care: per-class metrics on them are high-variance, and class weighting or oversampling should be evaluated.